NIOS Class 10 (211) Mathematics — Syllabus Bifurcation & Exam Blueprint

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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> **CRITICAL TIME-SAVING RULE FOR NOVEMBER 5 EXAM:**
> NIOS has officially bifurcated the syllabus into **TMA (40%)** and **Public Examination (60%)**. **DO NOT waste your precious time studying TMA chapters.** TMA chapters are evaluated ONLY in assignments and **will NOT be asked in the Public Examination**.

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1. What to Study vs What to Completely Skip

Module	Total Chapters	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)	Module Marks Weightage
Module 1: Algebra	7	• Ch 1: Number System • Ch 2: Exponents & Radicals • Ch 3: Algebraic Expressions	• **Ch 4: Special Product & Factorisation** • **Ch 5: Linear Equations** • **Ch 6: Quadratic Equations** • **Ch 7: Arithmetic Progression (AP)**	**20 Marks**
Module 2: Commercial Maths	2	• Ch 8: Percentage & Its Applications	• **Ch 9: Installment Buying**	**8 Marks**
Module 3: Geometry	10	• Ch 10: Lines & Angles • Ch 11: Congruence of Triangles • Ch 12: Concurrent Lines • Ch 13: Quadrilaterals • Ch 14: Similarity of Triangles • Ch 15: Circles	• **Ch 16: Angles in Circle & Cyclic Quads** • **Ch 17: Tangents, Secants & Properties** • **Ch 18: Constructions** • **Ch 19: Coordinate Geometry**	**25 Marks**
Module 4: Mensuration	2	• Ch 20: Perimeter & Area of Plane Figures	• **Ch 21: Surface Area and Volume**	**10 Marks**
Module 5: Trigonometry	2	• Ch 22: Introduction to Trigonometry	• **Ch 23: Trigonometric Ratios & Angles (Heights & Distances)**	**10 Marks**

Module	Total Chapters	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)	Module Marks Weightage
Module 6: Statistics & Probability	3	• Ch 24: Data & Their Representation	**• Ch 25: Measure of Central Tendency (Mean, Median)** **• Ch 26: Probability**	**12 Marks**
TOTAL	**26 Lessons**	**13 Lessons SKIPPED (Save 50% Time!)**	**13 High-Yield Lessons ONLY**	**85 Marks**

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2. Question Paper Pattern (85 Marks, 2½ Hours)

The exam paper contains **44 questions** divided into two sections:

Section A: Objective Questions — 42 Marks (49.4% of the Paper!)

- **Q1 to Q17 (17 Questions × 1 Mark = 17 Marks)**: Multiple Choice Questions (MCQs). Direct formula values, basic concepts.
- **Q18 to Q27 (10 Questions × 2 Marks = 20 Marks)**: Objective Type (each question has two 1-mark subparts: Fill in the blanks, True/False, Match the column).
- **Q28 (1 Question × 5 Marks = 5 Marks)**: Case Study / Context question with five 1-mark subparts.

> **Key Strategy**: Scoring 35+ out of these 42 objective marks immediately guarantees passing with flying colours!

Section B: Subjective Questions — 43 Marks

- **Q29 to Q37 (9 Questions × 2 Marks = 18 Marks)**: Very Short Answer (VSA). (4 have internal choices).
- **Q38 to Q42 (5 Questions × 3 Marks = 15 Marks)**: Short Answer (SA). (2 have internal choices).
- **Q43 to Q44 (2 Questions × 5 Marks = 10 Marks)**: Long Answer (LA). (Both have internal choices).

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3. The "Sure-Shot" 30 Marks Checklist

If you master just these 6 topics, you will easily secure 30–35 marks in Section B alone:

1. **Heights & Distances (Trigonometry) [5 Marks]**:
2. Always Q44 (Light House / Tower / Building with angles $30^\circ, 45^\circ, 60^\circ$).
3. **Graphical Solution of Linear Equations OR Circle Tangents Construction [5 Marks]**:
4. Always Q43 (Plot two straight lines on graph sheet to find intersection, or construct tangents to circle from external point).
5. **Median / Mean of Grouped Data [3 Marks]**:
6. Always Q41 or Q42 (Using the median formula $L + \frac{\frac{N}{2} - cf}{f} \times h$).
7. **Coordinate Geometry [3 to 5 Marks]**:
8. Distance formula, Mid-point formula, and Section formula dividing line segment.
9. **Arithmetic Progression (AP) [3 to 4 Marks]**:
10. Finding the n-th term ($a_n = a + (n-1)d$) and sum of n terms (S_n).
11. **Probability [3 Marks]**:
12. Simple problems on single/double dice, coin toss, or deck of 52 cards.